Fundamentals Chapter 3

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1. Consider the group $G = (\mathbb{Z}/p\mathbb{Z})^\times$ under multiplication where p is prime. The order of G is $p-1$. By Lagrange's theorem, the order of any subgroup of G divides $|G|$. Now, consider $a \in G$ and the subgroup $\langle a \rangle \leq G$ and let $|\langle a \rangle| = k$. We know that $a^k \equiv 1 \pmod{p}$, so

$$a^{p-1} \equiv (a^k)^{\frac{p-1}{k}} \equiv 1 \pmod{p}$$

as desired. Multiplying by a gives us $a^p \equiv a \pmod{p}$.

2. There is no group G and subgroup $H \leq G$ where $(aH)(bH) = abH$ for all $a, b \in G$. To see this, let $b = a^{-1}$. Then $(aH)(a^{-1}H) = \{ah_1a^{-1}h_2 : h_1, h_2 \in H\}$. By the condition, this is equal to H, which means that $ah_1a^{-1} \in H$, which implies that H is normal.

3. Since G is abelian, $ab = ba$ for all $a, b \in G$. So

$$aH = \{ah : h \in H\} = \{ha : h \in H\} = Ha,$$

which means H is normal.

5. Let $K = \ker(\phi)$. Then for all $k \in K$, $\phi(k) = e_G$. Also, since ϕ is a homomorphism,

$$\phi(ak) = \phi(a)\phi(k) = \phi(a)e_G = \phi(a),$$
$$\phi(ka) = \phi(k)\phi(a) = e_G\phi(a) = \phi(a)$$

for all $a \in H$. So $\phi(ak) = \phi(ka)$. This implies that, for all $k_1 \in K$, there exists $k_2 \in K$ such that $ak_1 = k_2a$. Therefore, $aK = Ka$ and K is a normal subgroup of H.

7. From problem 5, if there's a group H and a homomorphism $\phi : G \to H$ then $\ker(\phi)$ is normal.

Assume $K \leq G$ is normal. Now, define a map $\phi : G \to G/K$ by $\phi(g) = gK$ for all $g \in G$. This is a homomorphism because

$$\phi(g_1g_2) = (g_1g_2)K = (g_1K)(g_2K) = \phi(g_1)\phi(g_2)$$

for all $g_1, g_2 \in G$.

The kernel of ϕ is

$$\ker(\phi) = \{g \in G : \phi(g) = K\} = \{g \in G : gK = K\},$$

which simplifies to

$$\ker(\phi) = \{g \in G : g \in K\} = K,$$

so $\ker(\phi) = K$.

9a. We need to show that $g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x$ and $e \cdot x = x$ for all $g_1, g_2 \in G$ and $x \in G$. Looking at $g_1 \cdot (g_2 \cdot x)$, we get

$$\begin{aligned} g_1 \cdot (g_2 \cdot x) &= g_1 \cdot (g_2 x g_2^{-1}) \\ &= g_1 (g_2 x g_2^{-1}) g_1^{-1} \\ &= g_1 g_2 x g_2^{-1} g_1^{-1}. \end{aligned}$$

Also, $e \cdot x = exe^{-1} = exe = x$, so $\cdot$ defines a group action.

9b. Since $C(x) \subset G$, we need to show that $e \in C(x)$ and $g^{-1} \in C(x)$ for all $g \in C(x)$. Since $ge = eg$ for all $g \in G$, $e \in C(x)$.

To show that $g^{-1} \in C(x)$, we need to prove that $g^{-1}x = xa^{-1}$. Since $a \in C(x)$, we know that $ax = xa$, so

$$\begin{aligned} e &= (xa)(xa)^{-1} \\ &= (ax)(xa)^{-1} \\ &= axa^{-1}x^{-1}. \end{aligned}$$

Multiplying both sides by a^{-1} on the left and then x on the right, we get

$$\begin{aligned} a^{-1} &= xa^{-1}x^{-1} \\ a^{-1}x &= xa^{-1}, \end{aligned}$$

which means $a^{-1} \in C(x)$.

10a. Consider the set:

$$\begin{aligned} aZ(G) &= a\{x \in G : gx = xg\} \\ &= \{ax \in G : gx = xg\} \\ &= \{xa \in G : gx = xg\} \\ &= Z(G)a, \end{aligned}$$

so $Z(G)$ is normal.

10b. If $G/Z(G)$ is cyclic then there exists $gZ(G) \in G/Z(G)$ such that every coset in $G/Z(G)$ can be written as $(gZ(G))^k$ for some $k \in \mathbb{Z}$. So, for all $h \in G$,

$$hZ(G) = (gZ(G))^k = g^k Z(G)$$

for some $k \in \mathbb{Z}$. This means h and g^k are in the same coset of $Z(G)$ in G. This implies that $h(g^k)^{-1} \in Z(G)$ or $h = g^k x$ for some $x \in Z(G)$.

Now, let $a, b \in G$ be

$$a = g^{k_1} x_1, \quad b = g^{k_2} x_2.$$

Then

$$\begin{aligned} ab &= g^{k_1} x_1 g^{k_2} x_2 \\ &= g^{k_1} g^{k_2} x_1 x_2 \\ &= g^{k_1+k_2} x_1 x_2 \\ ba &= g^{k_2} x_2 g^{k_1} x_1 \\ &= g^{k_2} g^{k_1} x_2 x_1 \\ &= g^{k_1+k_2} x_1 x_2 \end{aligned}$$

so $ab = ba$ for all $a, b \in G$ and G is abelian.

11. To show that $|aH| = |bH|$ for any $a, b \in G$, we need to establish a bijection from aH to bH. We can map each $ah \in aH$ to $ba^{-1}ah = bh \in bH$. Since multiplication by ba^{-1} is unique, this is a bijection.

Now, we need to show that if $aH \neq bH$ then $aH \cap bH = \varnothing$. Assume that $aH \cap bH \neq \varnothing$. Then there exists $x \in G$ such that $x \in aH$ and $x \in bH$. This implies that $x = ah_1 = bh_2$, so $b^{-1}a = h_2 h_1^{-1}$, so $b^{-1}a \in H$. This implies that $a \in bH$ and $aH = bH$, which is a contradiction. So $aH \cap bH = \varnothing$.